University of Bologna, Dept of Physics and Astronomy.
Course Planetary Boundary Layer and Turbulent Diffusion,
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Report on Numerical Simulation Workshop

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11th June 2026

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1 Cintolesi

1.1 Introduction

Urban canyons are street-level cavities between buildings that govern local wind patterns and pollutant dispersion.

This report analyses the flow and scalar transport inside a square canyon using RANS simulations under isothermal and heated boundary conditions, where the atmospheric wind is driven by a constant free stream U, at the top of the domain. The pollutant is released with constant flux at the street level from a uniform area corresponding to the roadway. We call our simulations:

- isothermal
- heated

1.2 Analyses

When solar radiation hits the canyon, it unevenly heats the pavement and the building facades. This creates buoyancy forces (warm air rising because it is less dense) that interact with the prevailing wind flowing over the roofs. The impact depends entirely on which surface is hot, creating a competition between mechanical wind forces and thermal convection forces

In the isothermal case a single clockwise recirculation traps pollutants near the downwind façade (Figure 2). Heating that façade introduces a buoyancy-driven upward current that disrupts the recirculation and reduces near-road concentrations (Figure 4).

All residuals decrease monotonically below 10^{-5}, confirming convergence (Figure 1).

Results

Thermal forcing improves canyon ventilation and reduces peak pollutant concentration compared to the isothermal reference case. The pollutant remains at the bottom of the canyon due to the shape of the currents that does not favor recirculation of it. RANS simulations show that heating the downwind façade significantly alters canyon circulation and enhances pollutant dispersion, with direct implications for urban air quality management.

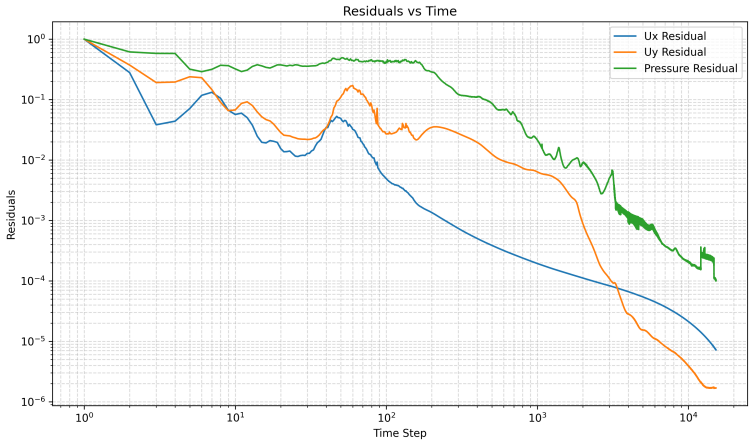


Figure 1: Solver residuals as a function of iteration number in the isothermal case. The resiudal drop quickier for U_x, U_y. The pressure decays slowly indicating that is more difficult to get it right. All three curves are trendily decreasing with few fluctuation.

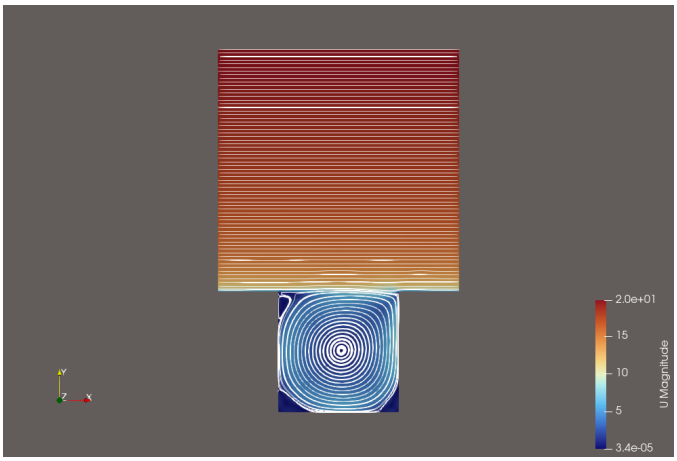


Figure 2: Speed profile for the isothermal case. The canyon is pretty much isolated with respect to the outside generating a big gradient in speed at the boundary of the free flow. A clockwise vortex is produced

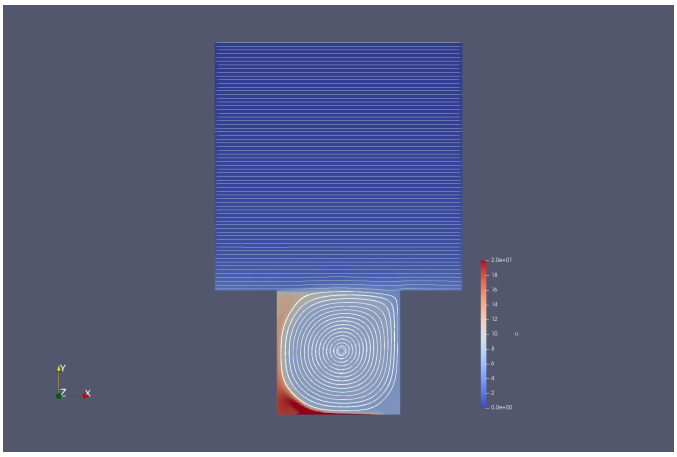


Figure 3: Pollutant concentration for the isothermal case. Pollutant is pushed by the vortex to the walls due to the clockwise vortex dragging it.

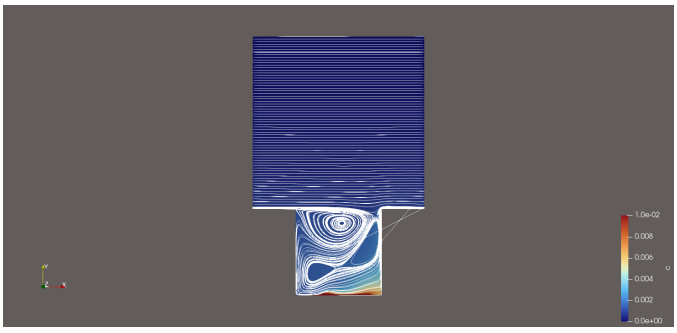


Figure 4: Pollutant concentration in the heated canyon. The streamline of the flow are more cahotic with resepct of those of 2 due to excess of energy poured in by heat. This, has the result of trapping the pollutant in the bottom of the canyon.

2 Marinacci

2.1 Introduction

The Sod shock tube is a standard benchmark for hydrodynamics codes, consisting of two regions of gas separated by a membrane at $x = 0$. The left state has density $\rho_L = 8\rho_R$ and pressure $P_L = 8P_R$, while both sides share the same total mass. Upon release, the discontinuity evolves into a rarefaction wave, a contact discontinuity, and a shock wave, for which an exact analytical solution exists. The test is reproduced here using the SPH solver GADGET-4 with $N = 64$ and $N = 512$ particles to assess convergence.

2.2 Analyses

The simulated profiles of density ρ, pressure P, velocity v, and specific entropy s are extracted at a fixed output time (1000 time steps). The two particle counts probe the effect of resolution on the ability of SPH to capture the sharp contact discontinuity and the shock front, which are known to be challenging for kernel-based methods.

2.3 Results

Figure 5 shows the four profiles for both resolutions. At $N = 64$ all features are qualitatively reproduced, but the contact discontinuity and the shock front are significantly broadened due to the large kernel support. Spurious entropy generation is visible at the contact surface, a well-known artefact of standard SPH. Increasing the particle count to $N = 512$ reduces the kernel length and sharpens all discontinuities; the density, pressure profiles, and the entropy jump is better localised. The velocity field improves accordingly, with the post-shock plateau converging to the expected value. Residual smoothing at the contact discontinuity persists even at $N = 512$, reflecting the intrinsic diffusivity of the SPH pressure formulation rather than a resolution effect.

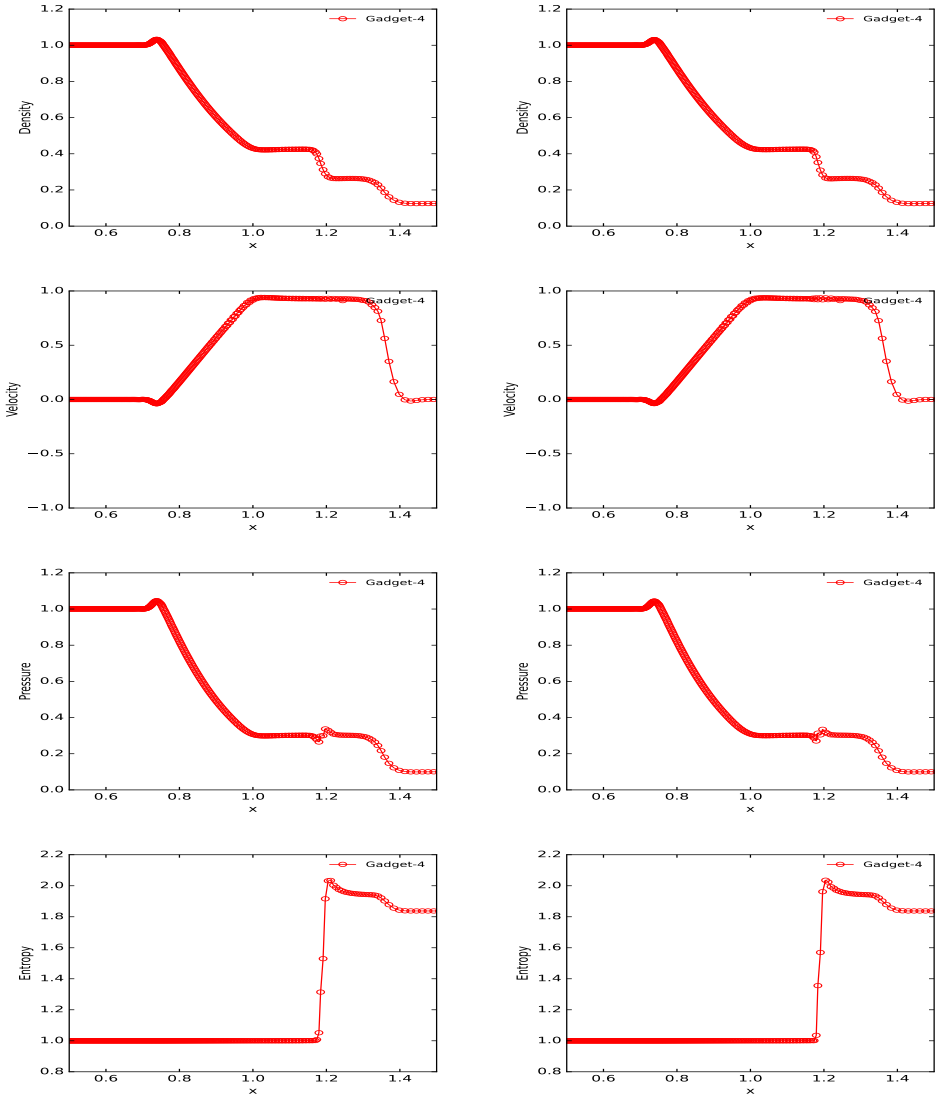


Figure 5: Sod shock tube profiles at fixed output time for $N = 64$ particles. (Left) non-SPH simulations, (Right) SPH. From top to bottom: density, velocity, pressure, and specific entropy. SPH should improve the description of the discontinuity. It is slightly visible in the pressure discontinuity where the curve is just a little smoother in the discontinuity.

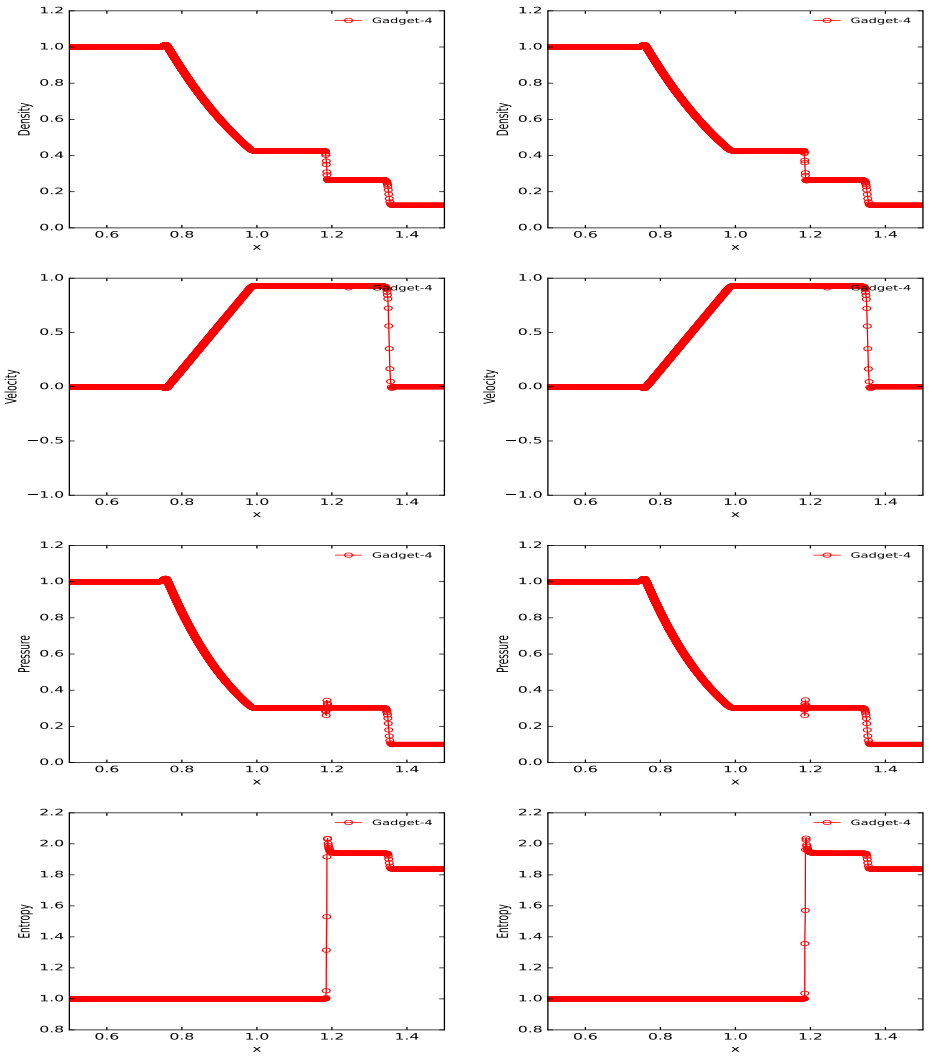


Figure 6: Sod shock tube profiles at fixed output time for $N = 512$ particles. (left) non SPH simulations, (right) SPH. From top-left to bottom-right: density, velocity, pressure,specific entropy. SPH should improve the description of the discontinuity. It is slightly visible in the pressure discontinuity where the curve is just little smoother in the discontinuity. They improve in sharpness relative to the 64 particle simulation 5

3 Calura

3.1 Introduction

In this project we use Ramses to repeat the experience about SOD already treated in the previous section and reproduce a 3D shock wave.

3.2 Analyses

The simulated profiles of density ρ, pressure P, are extracted at a fixed output time (1000 time steps). The two particle counts probe the effect of resolution on the ability of SPH to capture the sharp contact discontinuity and the shock front, which are known to be challenging for kernel-based methods in the SOD case. In the 3D explosion test wealso show velocity v, in all three axes.

3.3 Results

3.3.1 SOD

The SOD run was ran with the standard SOD parameter, as in the case of 5 in the previous section. We used 256 particles in this case. The simulation employs the HLLC Riemann solver, a second-order reconstruction scheme. As the system evolves, the initial discontinuity decomposes into the classical Riemann-wave structure composed of a left-propagating rarefaction wave, a contact discontinuity, and a right-propagating shock front. The AMR hierarchy automatically concentrates resolution in the vicinity of these features, allowing steep gradients to be captured while maintaining a comparatively coarse discretisation in smooth regions. Compared with uniform-grid methods of equivalent effective resolution, the adaptive strategy significantly reduces the computational cost while preserving the sharpness of the hydrodynamic structures.

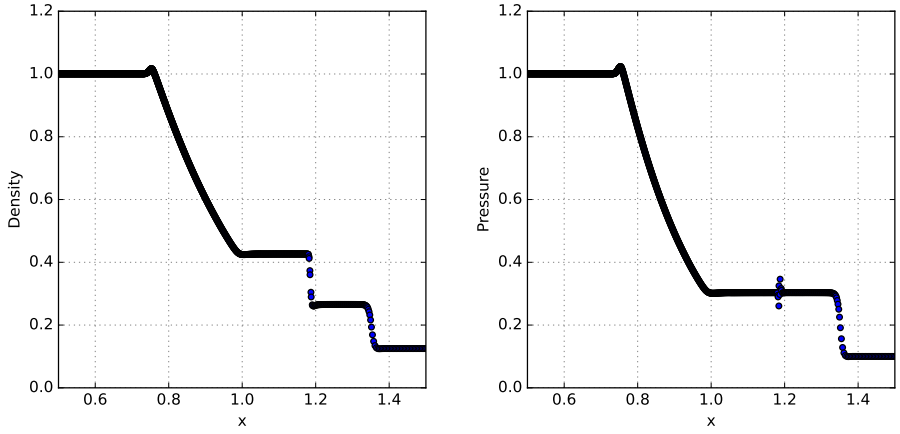


Figure 7: Density (left) and pressure (right) profile at last time step simulated in 1D sod. The simulation is not perfectly reproducing the discontinuity, however it handles them in a decent way, suggesting that resolution and number of particles used are adequate to extract the principal features of the system, even though the pressure is not completely converged to the plateau typical of the solution after the first drop.

3.3.2 3D Explosion

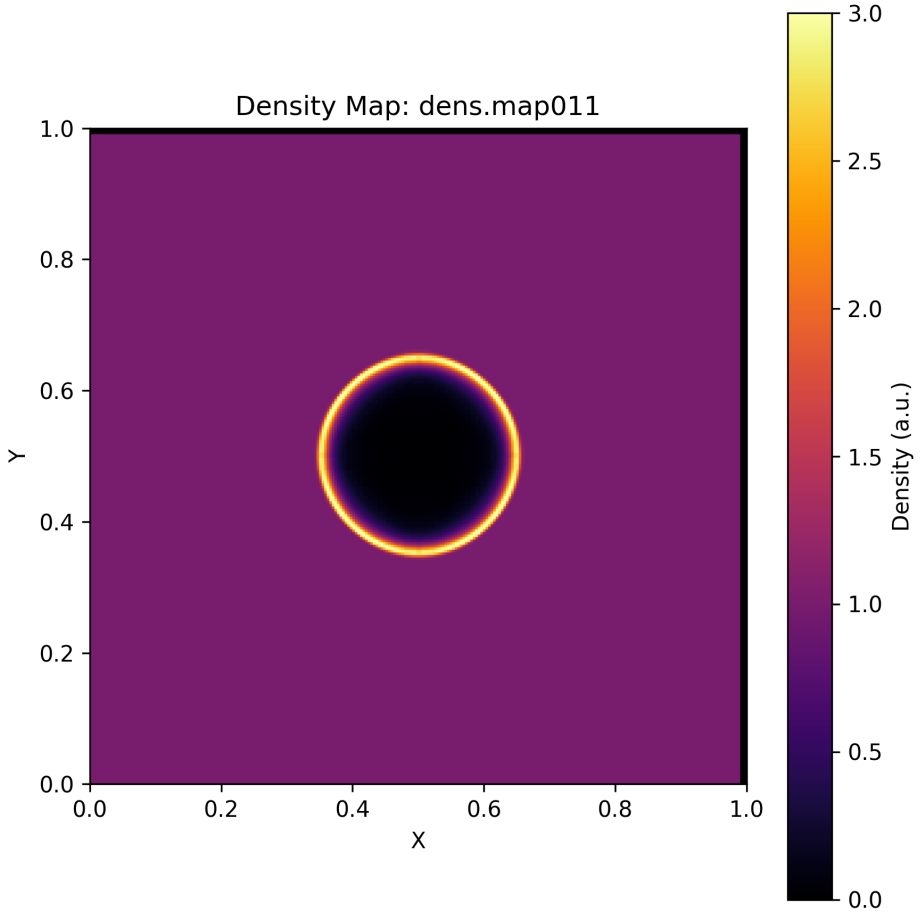


Figure 8: Density profile at last time step simulated. In my simulation I did not reach the boundary so I do not have reflection from them.

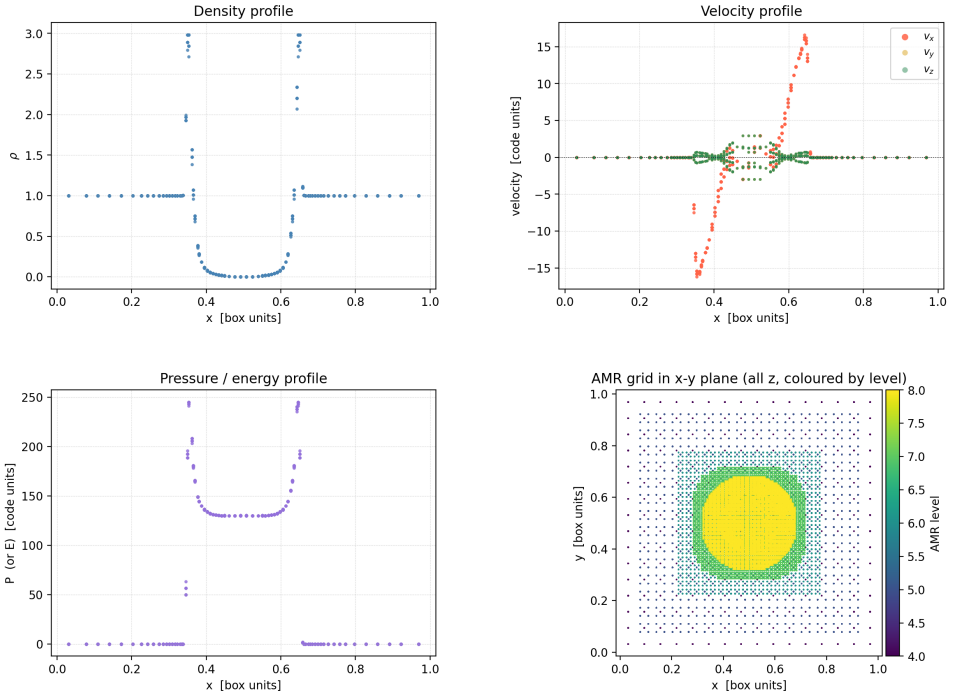


Figure 9: On the top left the density profile in the x direction fixing the plane z = 0, y = 0. It is clear the discontinuity front propagation. On the right the speed profile of the front in all three directions. (Bottom left) The discontinuity is a feature also of the pressure profile that drop to 0 after the front wave. (Bottom right) Visualization of the front wave on y plane with the visualization of AMR levels on top. Essentially same plot of 8, done to check that I was doing the right thing.

4 Aknowledgement

I used AI to produce the plot and in improving the english of the presentation.